

LESSON 2-9 • SESSION 1 • PRACTICE • FORM B

Finding Mean Absolute Deviation (MAD)

Learning target – I can find the mean absolute deviation of a data set and explain what it measures. • 6.SP.B.5c

REMEMBER

- **MAD** (mean absolute deviation) = the mean of the distances from the mean.
- Steps: find the **mean**, find each value's **distance** from it (never negative), then **average** the distances.
- If all values match, the MAD is 0. A **small** MAD means values sit close to the mean.

WORKED EXAMPLE

Four plants grew 3, 7, 8 and 10 cm. Find the MAD.

Step 1 Mean: $28 \div 4 = 7$.**Step 2** Distances from 7: 4, 0, 1, 3.**Step 3** Sum = 8, and $8 \div 4 = 2$.**Answer** **MAD = 2 cm**

WORDS mean absolute deviation (*desviación media absoluta*) • deviation (*desviación*) • absolute value (*valor absoluto*) • spread (*dispersión*)

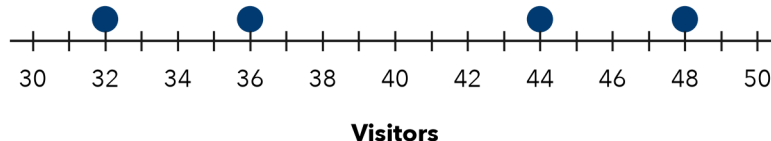
PRACTICE

show your work • label units

1. **Spelling Quiz.** The mean of 11, 14, 18, 22 and 25 is 18. Complete the table. Then find the MAD: _____

Value	11	14	18	22	25
Distance from 18					

2. **Gym Visits.** The dot plot shows the visitors to a gym on four evenings.



- a. Mean: _____ visitors
- b. MAD: _____ visitors
3. **Afternoon Temps.** Four afternoon temperatures were 61, 64, 68 and 71°F. Find the MAD.

Hint Find the mean first. Measure each distance from it.

Show your work.

Answer _____ °F

4. **All the Same.** Four values are 40, 40, 40, 40. Kenji says the MAD cannot be found. Is he right?

Hint Find each value's distance from the mean.

Show your work.

MAD _____

5. **Reading a MAD.** Six bus rides have a mean of 30 minutes and a MAD of 2 minutes. What does the MAD tell a rider?

Show your work.

Meaning _____

- 6. Homework Pages.** Five students turned in 10, 16, 22, 24 and 28 pages. Find the mean and the MAD.

Hint Measure distances from the mean, not from 0.

Show your work.

Mean, MAD _____

- 7. Missing Distance.** Four values have mean 60 and MAD 6. Three distances from the mean are 3, 7 and 5. Find the fourth distance.

Hint $MAD \times 4$ gives the total of the distances.

Show your work.

Distance _____

- 8. Two Groups.** Free throws made – Group 1: 5, 7, 9, 11. Group 2: 2, 8, 9, 13. Both means are 8.

a. MAD of Group 1: _____

b. MAD of Group 2: _____

c. What does the MAD tell you that the mean does not? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Tobias's Work.** For 26, 28, 34, 36 (mean 31), Tobias added -5 , -3 , $+3$, $+5$ and reported a MAD of 0. Explain his mistake and find the MAD.

TEST PRACTICE

choose the best answer

- 10.** A deviation is the distance between a value and the –

A mean

B median

C greatest value

D range

- 11.** A smaller MAD means the data is –

A more spread out

B larger

C closer to the mean

D negative

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

28	0	5.6	3	4.8	3.5	4.4	9	6	14
X	R	A	E	Z	E	A	G	V	Q

#1	#2b	#3	#4	#6	#7	#8b
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____